

Thawsif Chowdhury

Jamaica, NY 11433 • +1 (516) 557-5859 • thawsifchowdhury@gmail.com • [LinkedIn](#)

EDUCATION

Lehigh University, Bethlehem, PA

Graduation: May 2029

Bachelor of Science in Electrical Engineering

GPA: 3.96

Related Coursework: Multivariable Calculus, Electricity and Magnetism, Engineering Principles

Personal Projects

Automated Irrigation System

June 2024 – July 2024

- Designed a controlled irrigation system using Arduino UNO for maintaining optimum soil moisture levels
- Programmed in C++ to display real-time analog data of moisture levels using a soil sensor
- Implemented a threshold logic (500 ADC) which activated the motor via a relay module to water plants

Mechanical Arm

March 2024 – April 2024

- Designed a 5-fingered mechanical hand controlled using an Arduino UNO
- Implemented servo motors with strings that connected to the fingers
- Programmed the control software in Arduino IDE (C++) to rotate servo motors in response to keyboard inputs for finger articulation

EXPERIENCE

Software Engineering Intern

June 2026 – Present

ThinkNeuro - Virtual

- Manage version control using Git fundamentals such as merging, pulling, branching, committing to prevent code conflicts among collaborators

Global Social Impact Fellow - PureSpaces

January 2026 – Present

Lehigh Creative Inquiry, Bethlehem, PA

- Coordinating an international partnership between 3 universities to implement low-cost, high efficiency air purifiers in Almaty, Kazakhstan targeting a \$10.5B health crisis caused by extreme air pollution
- Lead the experimental validation of PureAura filtration systems, executing 9+ trials using Bunsen burners and IQ Air monitors to analyze 1-minute interval CSV data to ensure PM 2.5 stability

America Reads America Counts Tutor

September 2025 – May 2026

Lehigh Community Service Office, Bethlehem, PA

- Tutored 40+ 8th-grade students in Science, English, Math to prepare them for high school coursework
- Dedicated 3 hours per week across 3 sessions to clarify 10+ core concepts in Science, English, Math

Engineering Student Researcher

July 2024 – August 2024

Vaughn College, Flushing, NY

- Engineered and trained a Convolutional Neural Network (CNN) model for handwritten digit recognition and a Long Short-Term Memory (LSTM) model for performing sentiment analysis on a dataset of 50,000 IMDb movie reviews
- Implemented a machine learning system using a Linear Regression model for analyzing housing prices
- Engineered an AI-powered voice recognition program using Python to control game movements
- Developed a research project on AI voice recognition in games like Pac-Man

AP Calculus Tutor and Grader

December 2023 – June 2024

Hillcrest High School, Jamaica, NY

- Dedicated 80+ hours to mentoring 10+ peers in core Calculus I concepts, such as derivatives and integrals, through one-on-one tutoring sessions, successfully increasing the passing rate by 11%
- Graded 100+ assignments and exam papers

PROFESSIONAL ORGANIZATION

Treasurer | Institute of Electrical and Electronics Engineers, Lehigh University

September 2025 – Present

- Actively participating in technical workshops, such as a MATLAB session, while collaborating with peers

SKILLS

Programming Languages: Python, Java (Basics), Arduino (C++)

Technical Skills: Arduino UNO, Tinkercad, LTSpice, Git, GitHub, MS Office Suite, Google Suite